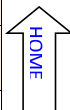


Savitribai Phule Pune University, Pune Third Year Information Technology (2019 Course) 314452: Data Science and Big Data Analytics		
Teaching Scheme:	Credit Scheme:	Examination Scheme:
Theory (TH) :03 Hrs/week	03 Credit	Mid_Semester : 30 Marks End_Semester : 70 Marks
Prerequisites: 1. Engineering and discrete mathematics. 2. Database Management Systems, Data warehousing and Data mining. 3. Programming skill.		
Companion Course: 1. Machine Learning 2. Advance Database Management		
Course Objectives: 1. To introduce basic need of Big Data and Data science to handle huge amount of data. 2. To understand the basic mathematics behind the Big data. 3. To understand the different Big data processing technologies. 4. To understand and apply the Analytical concept of Big data using Python. 5. To visualize the Big Data using different tools. 6. To understand the application and impact of Big Data.		
Course Outcomes: On completion of the course, students will be able to– CO1: Understand Big Data primitives. CO2: Learn and apply different mathematical models for Big Data. CO3: Demonstrate Big Data learning skills by developing industry or research applications. CO4: Analyze and apply each learning model comes from a different algorithmic approach and it will perform differently under different datasets. CO5: Understand, apply and analyze needs, challenges and techniques for big data visualization. CO6: Learn different programming platforms for big data analytics.		
COURSE CONTENTS		
Unit I	INTRODUCTION: DATA SCIENCE AND BIG DATA	(06 Hrs)
Introduction to Data science and Big Data , Defining Data science and Big Data, Big Data examples, Data Explosion : Data Volume, Data Variety, Data Velocity and Veracity. Big data infrastructure and challenges, Big Data Processing Architectures : Data Warehouse, Re-Engineering the Data Warehouse, shared everything and shared nothing architecture, Big data learning approaches. Data Science – The Big Picture: Relation between AI, Statistical Learning, Machine Learning , Data Mining and Big Data Analytics.		



Mapping of Course Outcomes for Unit I	CO1	
Unit II	MATHEMATICAL FOUNDATION OF BIG DATA	(07 Hrs)
Probability: Random Variables and Joint Probability, Conditional Probability and concept of Markov chains, Tail bounds, Markov chains and random walks, Pair-wise independence and universal hashing, Approximate counting, Approximate median. Data Streaming Models and Statistical Methods: Flajolet Martin algorithm, Distance Sampling and Random Projections, Bloom filters, Mode, Variance, standard deviation, Correlation analysis and Analysis of Variance.		
Mapping of Course Outcomes for Unit II	CO2	
Unit III	BIG DATA PROCESSING	(06 Hrs)
Big Data Analytics- Ecosystem and Technologies, Introduction to Google file system, Hadoop Architecture, Hadoop Storage: HDFS, Common Hadoop Shell commands, Anatomy of File Write and Read, NameNode, Secondary NameNode, and DataNode, Hadoop MapReduce paradigm, Map Reduce tasks, Job, Task trackers - Cluster Setup – SSH & Hadoop Configuration, Introduction to NOSQL, Textual ETL processing.		
Mapping of Course Outcomes for Unit III	CO3	
Unit IV	BIG DATA ANALYTICS	(06 Hrs)
Big Data Analytics- Architecture and Life Cycle, Types of analysis, Analytical approaches, Data Analytics with Mathematical manipulations, Data Ingestion from different sources (CSV, JSON, html, Excel, mongoDB, mysql, sqlite), Data cleaning, Handling missing values, data imputation, Data transformation, Data Standardization, handling categorical data with 2 and more categories, statistical and graphical analysis methods, Hive Data Analytics.		
Mapping of Course Outcomes for Unit IV	CO4	
Unit V	BIG DATA VISUALIZATION	(06 Hrs)
Introduction to Data visualization, Challenges to Big data visualization, Conventional datavisualization tools, Techniques for visual data representations, Types of data visualization, Visualizing Big Data, Tools used in data visualization, Propriety Data Visualization tools, Open – source data visualization tools, Case Study: Analysis of a business problem of Zomato using visualization, Analytical techniques used in Big data visualization, Data Visualization using Tableau Introduction to: Candela, D3.js, Google Chart API		

Mapping of Course Outcomes for Unit V	CO5	
Unit VI	BIG DATA TECHNOLOGIES APPLICATION AND IMPACT	(05 Hrs)
Social media analytics, Text mining, Mobile analytics, Data analytics life cycle of case studies, Organizational impact, understanding decision theory, creating big data strategy, big data value creation drivers, Michael Porter’s valuation creation models, Big data user experience ramifications, Identifying big data use cases, Big Data Analytics Challenges and Research directions.		
Mapping of Course Outcomes for Unit VI	CO6	
Text Books:		
1. Krish Krishnan, Data warehousing in the age of Big Data, Elsevier, ISBN: 9780124058910, 1 st Edition.		
2. DT Editorial Services, Big Data, Black Book, DT Editorial Services, ISBN: 9789351197577, 2016Edition.		
Reference Books:		
1. Mitzenmacher and Upfal, Probability and Computing: Randomized Algorithms and ProbabilisticAnalysis, Cambridge University press, ISBN :521835402 .		
2. Dana Ron, Algorithmic and Analysis Techniques in Property Testing, School of EE.		
3. Graham Cormode, Minos Garofalakis, Peter J. Haas and Chris Jermaine, Synopses for Massive Data: Samples, Histograms, Wavelets, Sketches, Foundation and trends in databases, ISBN:10.1561/1900000004.		
4. Alex Holmes, Hadoop in practice, Dreamtech press, ISBN:9781617292224.		
5. AmbigaDhiraj, Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends forToday’s Business, Wiely CIO Series.		
6. ArvindSathi, Big Data Analytics: Disruptive Technologies for Changing the Game, IBMCorporation, ISBN:978-1-58347-380-1.		
7. EMC Education Services, Data Science and Big Data Analytics- Discovering, analyzing Visualizingand Presenting Data.		
8. Li Chen, Zhixun Su, Bo Jiang, Mathematical Problems in Data Science, Springer, ISBN :978-3-319-25127-1.		
9. Philip Kromer and Russell Jurney, Big Data for chips, O’Reilly, ISBN :9789352132447.		
10. EMC Education services, Data Science and Big Data Analytics, EMC2 Wiley, ISBN :978812655653-		
11. Mueller Massaron, Python for Data science, Wiley, ISBN :9788126557394.		
12. EMC Education Services, Data Science and Big Data Analytics, Wiley India, ISBN:9788126556533		
13. Benoy Antony, Konstantin Boudnik, Cheryl Adams,,Professional Hadoop, Wiley India,ISBN :9788126563029		
14. Judith Hurwitz, Alan Nugent, Big Data For Dummies, Wiley India, ISBN : 9788126543281		

E Books / E Learning References :

1. Zomato dataset Link: <https://www.kaggle.com/shrutimehta/zomato-restaurants-data>
2. Link for dataset: <https://www.kaggle.com/tanmoyie/us-graduate-schools-admission-parameters>